

Probabilistic Recognition of Almost Primes by Machine Learning

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Appendix

A Parameters of the Integer-Vector Encodings

This appendix provides the implementation details and supplementary experimental results supporting the main text. Section A records the constants defining the two integer-vector encodings. Sections B–D summarize the dataset composition, detailed confusion matrix statistics, and sensitivity analyses with respect to the remainder dimension and encoding choice.

Both integer-vector encodings use the base

$$M = 2^{44}.$$

Digit encoding e_d . The encoding e_d applies no additional transformation; its d coordinates are the base- M digits of n itself.

Hash encoding h_d . The encoding h_d first computes the linear transformation

$$A_2 n, \quad A_2 = 2^{42} + 1,$$

and then takes the remainders modulo the first d of the four primes

$$\begin{aligned} p_1 &= 2^{45} + 75, & p_2 &= p_1 + 39 \cdot 2^{38}, \\ p_3 &= 2^{46} + 15, & p_4 &= 2^{46} - 2^{44} + 55. \end{aligned}$$

The values p_1, p_2, p_3, p_4 are distinct primes lying in $[2^{45}, 2^{47}]$, and hence each remainder is exactly representable in IEEE 754 double precision.

B Dataset Composition

Tables 1–4 summarize the natural target-class distributions at the training scale 2^{44} and the three larger testing scales 2^{88} , 2^{132} , and 2^{176} . For each $r \in \{2, \dots, 8\}$, the target density π_r is calculated relative to the integers satisfying $2 \leq \Omega(n) \leq 8$.

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r	π_r	No. of target integers	Training targets	Test targets
2	12.86%	499,649	399,719	99,930
3	21.48%	834,714	667,771	166,943
4	23.04%	895,319	716,255	179,064
5	18.59%	722,273	577,818	144,455
6	12.50%	485,781	388,625	97,156
7	7.44%	289,125	231,300	57,825
8	4.10%	159,424	127,539	31,885

Table 1: Target-class counts at the training scale 2^{44} . The full sampling window contains $N_{\text{total}} = 2^{22} = 4,194,304$ integers. Under stratified five-fold cross-validation, each fold uses 80% of the target class for training and 20% for testing. The columns “Training targets” and “Test targets” report the corresponding per-fold counts. Here π_r denotes the target density among integers satisfying $2 \leq \Omega(n) \leq 8$.

r	π_r	No. of target integers
2	7.84%	298,901
3	16.09%	613,235
4	21.12%	804,881
5	20.62%	786,002
6	16.35%	623,177
7	11.15%	424,844
8	6.83%	260,215

Table 2: Target-class counts at the test scale 2^{88} . The full sampling window contains $N_{\text{total}} = 2^{22} = 4,194,304$ integers. The entire retained dataset is used for evaluation; no training occurs at this scale. Here π_r denotes the target density among integers satisfying $2 \leq \Omega(n) \leq 8$.

C Additional Classification Results

Tables 5 and 6 report the complete confusion-matrix statistics underlying the P_4 values presented in the main text. These tables allow readers to examine the trade-offs among precision, recall, specificity, and negative predictive value for each classifier.

D Sensitivity Analyses

Tables 7 and 8 examine the robustness of the experimental results with respect to two design choices: the number of remainder features j and the choice of integer-vector encoding.

r	π_r	No. of target integers
2	5.89%	218,967
3	13.26%	492,807
4	19.41%	721,193
5	20.96%	778,757
6	18.20%	676,190
7	13.44%	499,356
8	8.82%	327,619

Table 3: Target-class counts at the test scale 2^{132} . The full sampling window contains $N_{\text{total}} = 2^{22} = 4,194,304$ integers. The entire retained dataset is used for evaluation; no training occurs at this scale. Here π_r denotes the target density among integers satisfying $2 \leq \Omega(n) \leq 8$.

r	π_r	No. of target integers
2	4.80%	174,028
3	11.58%	419,746
4	18.03%	653,767
5	20.87%	756,955
6	19.28%	699,070
7	15.05%	545,747
8	10.40%	377,016

Table 4: Target-class counts at the test scale 2^{176} . The full sampling window contains $N_{\text{total}} = 2^{22} = 4,194,304$ integers. The entire retained dataset is used for evaluation; no training occurs at this scale. Here π_r denotes the target density among integers satisfying $2 \leq \Omega(n) \leq 8$.

Model	r	Prec ₋₁	NPV	Recall ₋₁	Specificity	Accuracy
LR	2	0.2262	0.9059	0.4595	0.7681	0.7284
	3	0.3045	0.8459	0.5721	0.6425	0.6274
	4	0.2661	0.8065	0.5865	0.5158	0.5321
	5	0.2365	0.8504	0.5305	0.6091	0.5945
	6	0.2099	0.9175	0.5605	0.6987	0.6814
	7	0.1938	0.9519	0.4705	0.8427	0.8150
	8	0.2746	0.9709	0.3244	0.9633	0.9371
	8	0.2746	0.9709	0.3244	0.9633	0.9371
RF	2	0.3433	0.9456	0.6855	0.8065	0.7910
	3	0.3534	0.8813	0.6735	0.6629	0.6651
	4	0.3242	0.8306	0.5542	0.6542	0.6312
	5	0.3225	0.8597	0.4336	0.7921	0.7254
	6	0.3050	0.9201	0.4890	0.8408	0.7969
	7	0.3654	0.9407	0.2421	0.9662	0.9123
	8	0.4235	0.9736	0.3799	0.9779	0.9533
	8	0.4235	0.9736	0.3799	0.9779	0.9533

Table 5: Same-scale confusion-matrix statistics at the training/test scale 2^{44} using encoding h_4 with $j = 8$. Class -1 is the target class ($\Omega(n) = r$), while class $+1$ is the background class. NPV denotes negative predictive value. Specificity is the recall for the background class. See Section 6.1 for the corresponding discussion of same-scale performance.

Test scale	Model	r	Prec ₋₁	NPV	Recall ₋₁	Specificity	Accuracy
2^{88}	LR	2	0.0995	0.9325	0.4323	0.6672	0.6488
		3	0.1930	0.8654	0.5398	0.5674	0.5629
		4	0.2371	0.8126	0.5376	0.5370	0.5371
		5	0.1850	0.7796	0.3589	0.5892	0.5417
		6	0.1944	0.8521	0.3994	0.6764	0.6311
		7	0.2171	0.9082	0.3060	0.8615	0.7996
		8	0.1697	0.9340	0.0542	0.9806	0.9173
	RF	2	0.1746	0.9345	0.2634	0.8940	0.8446
		3	0.2513	0.9021	0.6413	0.6336	0.6349
		4	0.1920	0.7851	0.1460	0.8355	0.6899
		5	0.1886	0.7916	0.0990	0.8893	0.7263
		6	0.2623	0.8655	0.3641	0.7998	0.7286
		7	0.2227	0.8968	0.1387	0.9393	0.8500
		8	0.2454	0.9350	0.0656	0.9852	0.9224
2^{132}	LR	2	0.0905	0.9547	0.4625	0.7088	0.6943
		3	0.1651	0.8915	0.5316	0.5888	0.5812
		4	0.2244	0.8371	0.5870	0.5112	0.5259
		5	0.1988	0.7829	0.3872	0.5861	0.5444
		6	0.2161	0.8354	0.4009	0.6765	0.6263
		7	0.2400	0.8863	0.2929	0.8559	0.7802
		8	0.2730	0.9155	0.0613	0.9842	0.9028
	RF	2	0.1273	0.9582	0.4338	0.8137	0.7913
		3	0.2140	0.9391	0.7561	0.5752	0.5992
		4	0.1952	0.8060	0.1251	0.8757	0.7300
		5	0.1799	0.7861	0.1070	0.8705	0.7105
		6	0.2598	0.8358	0.2665	0.8310	0.7283
		7	0.2438	0.8734	0.1211	0.9417	0.8314
		8	0.4576	0.9157	0.0538	0.9938	0.9109
2^{176}	LR	2	0.0603	0.9586	0.4366	0.6571	0.6465
		3	0.1471	0.9124	0.6007	0.5441	0.5506
		4	0.2134	0.8474	0.5391	0.5630	0.5587
		5	0.1897	0.7789	0.3568	0.5979	0.5476
		6	0.2143	0.8163	0.3292	0.7117	0.6380
		7	0.2656	0.8653	0.2133	0.8955	0.7928
		8	0.3327	0.8994	0.0465	0.9892	0.8912
	RF	2	0.1223	0.9781	0.6630	0.7601	0.7555
		3	0.1891	0.9587	0.8232	0.5378	0.5708
		4	0.1781	0.8189	0.2779	0.7179	0.6386
		5	0.1611	0.7834	0.1098	0.8491	0.6948
		6	0.2494	0.8202	0.2405	0.8272	0.7141
		7	0.3209	0.8579	0.0997	0.9626	0.8327
		8	0.4667	0.9008	0.0579	0.9923	0.8952

Table 6: Cross-scale confusion-matrix statistics. The classifiers are trained at the scale 2^{44} and applied, without retraining, at the testing scales 2^{88} , 2^{132} , and 2^{176} , using encoding h_4 with $j = 8$. Class -1 is the target class ($\Omega(n) = r$), while class $+1$ is the background class. NPV denotes negative predictive value. Specificity is the recall⁴ for the background class. See Section 6.2 for the corresponding discussion of cross-scale performance.

j	Model	$r=2$	$r=3$	$r=4$	$r=5$	$r=6$	$r=7$	$r=8$
$j = 4$	RG	0.224	0.337	0.355	0.303	0.219	0.138	0.079
	LR	0.4445	0.5147	0.4628	0.4480	0.4412	0.4199	0.4554
	RF	0.6010	0.5731	0.5248	0.5187	0.5298	0.5616	0.5851
$j = 6$	RG	0.224	0.337	0.355	0.303	0.219	0.138	0.079
	LR	0.4444	0.5148	0.4629	0.4479	0.4411	0.4201	0.4552
	RF	0.6044	0.5760	0.5258	0.5124	0.5278	0.5104	0.5765
$j = 8$	RG	0.224	0.337	0.355	0.303	0.219	0.138	0.079
	LR	0.4443	0.5147	0.4629	0.4479	0.4411	0.4201	0.4549
	RF	0.5998	0.5749	0.5248	0.5106	0.5264	0.4462	0.5679
$j = 16$	RG	0.224	0.337	0.355	0.303	0.219	0.138	0.079
	LR	0.4444	0.5148	0.4629	0.4479	0.4411	0.4201	0.4550
	RF	0.5906	0.5704	0.5180	0.4921	0.5243	0.4329	0.5612

Table 7: P_4 values across remainder dimension j at the same-scale testing scale 2^{44} using encoding h_4 . The row labeled P_4^{RG} reports the natural random-guessing baseline defined in Section 5 and is independent of j ; the same value is therefore repeated in each panel for reference. The values of P_4^{LR} are essentially invariant in j , whereas P_4^{RF} is generally stable except at $r = 7$. This table supports the choice of $j = 8$ for the principal experiments discussed in Section 6.1.

Test scale	Encoding	Model	$r = 2$	$r = 3$	$r = 4$	$r = 5$	$r = 6$	$r = 7$	$r = 8$
2^{44}	RG		0.224	0.337	0.355	0.303	0.219	0.138	0.079
	h_4	LR	0.4443	0.5147	0.4629	0.4479	0.4411	0.4201	0.4549
		RF	0.5998	0.5749	0.5248	0.5106	0.5264	0.4462	0.5679
	e_4	LR	0.4164	0.4847	0.4643	0.4006	0.4321	0.4199	0.3652
		RF	0.6029	0.5717	0.5158	0.5040	0.4876	0.4999	0.4780
2^{88}	RG		0.145	0.270	0.333	0.327	0.274	0.198	0.127
	h_4	LR	0.2679	0.4020	0.4362	0.3581	0.3883	0.3946	0.1512
		RF	0.3415	0.4863	0.2753	0.2248	0.4462	0.2882	0.1870
	e_4	LR	0.3149	0.4150	0.4531	0.3728	0.3985	0.4266	0.4053
		RF	0.4681	0.5041	0.4681	0.4163	0.4289	0.4610	0.4161
2^{132}	RG		0.111	0.230	0.313	0.331	0.298	0.233	0.161
	h_4	LR	0.2552	0.3718	0.4296	0.3775	0.4083	0.4050	0.1812
		RF	0.3217	0.4546	0.2581	0.2309	0.4000	0.2746	0.1749
	e_4	LR	0.2586	0.3690	0.4353	0.3602	0.3740	0.4157	0.4078
		RF	0.3896	0.4522	0.4317	0.3702	0.3858	0.4187	0.3564
2^{176}	RG		0.091	0.205	0.296	0.330	0.311	0.256	0.186
	h_4	LR	0.1866	0.3510	0.4212	0.3626	0.3871	0.3729	0.1502
		RF	0.3327	0.4252	0.3382	0.2251	0.3775	0.2606	0.1858
	e_4	LR	0.2204	0.3377	0.4182	0.3515	0.3572	0.4041	0.4043
		RF	0.3355	0.4149	0.4029	0.3402	0.3565	0.3872	0.3141

Table 8: P_4 values by encoding, model, and testing scale for $j = 8$, with training performed at the scale 2^{44} . Encoding h_4 is used throughout the principal experimental results, whereas e_4 is included here to evaluate the robustness of the classifiers with respect to the choice of integer-vector encoding. The same-scale results at 2^{44} are included for reference. The row labeled P_4^{RG} reports the natural random-guessing baseline at each testing scale and does not depend on the encoding. Cells in which e_4 improves on h_4 by at least 0.10 may be highlighted in bold. This table supports the encoding comparison discussed in Section 6.2.